

SoulScope Inference Rule Registry v0.1

Status: Canonical scientific implementation companion — architecture final, numerical calibration pending

Governing authority: SoulScope Canon v1.3

Upstream: Acoustic Parameter Registry v0.1; Evidence Marker Registry v0.1

Downstream: Constellation Dimension Registry v0.1; State Registry v0.1

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NORTH STAR

SoulScope shall combine evidence without hiding disagreement, inventing missing information, or converting engineering assumptions into scientific certainty.

1. Purpose

The Inference Rule Registry defines how qualified Evidence Markers become:

1. Dimension estimates;
2. uncertainty intervals;
3. confidence;
4. evidence coverage;
5. contradiction;
6. coherence;
7. baseline trust;
8. temporal/contextual momentum;
9. resolved, unresolved, suppressed, or research-only decisions.

The registry governs:

Evidence → Dimension inference → Decision Ledger.

It does not define canonical states.

State selection occurs only after Dimension inference is complete.

2. Core principle

SoulScope does not use:

single acoustic feature → single semantic score.

It uses:

Measurements → Evidence Markers → independent Evidence families → Dimension posterior

while preserving:

quality, coverage, contradiction, confounds, baseline trust, missingness.

3. Rule types

Every active inference rule has one canonical type.

RULE_HARD_GATE

Determines whether inference is scientifically permitted.

Examples:

- sufficient valid speech;
- required prompt available;
- required Evidence family available;
- extractor compatibility;
- active construct model exists.

Failure may force unresolved.

RULE_SUPPORT

Defines Evidence that increases support for a candidate Dimension direction or region.

RULE_CONTRADICTION

Defines Evidence incompatible with or materially weakening a candidate interpretation.

RULE_CONFOUND

Defines conditions that reduce confidence, invalidate Evidence, or block inference.

RULE_MISSINGNESS

Defines consequences of absent required information.

RULE_REFERENCE

Defines which comparison basis is permitted.

RULE_AGGREGATION

Defines how multiple Evidence observations are combined.

RULE_CALIBRATION

Defines mapping from evidence space to posterior probability or tendency.

RULE_ABSTENTION

Determines when SoulScope must not publish a resolved Dimension.

RULE_PUBLICATION

Determines whether a technically computed result is permitted to appear in the production Semantic Result.

4. Permanent rule object

Every inference rule SHALL contain:

```
rule_id
rule_version

rule_type
dimension_id

scientific_status
```

```
required_prompt_scope[]
required_reference_scope[]
required_evidence_families[]
optional_evidence_families[]

required_markers[]
support_markers[]
contradiction_markers[]

hard_confounds[]
soft_confounds[]

minimum_coverage_policy
independence_policy

aggregation_method
calibration_model_id

abstention_conditions[]
publication_conditions[]

rationale
validation_status

created_at
approved_at
deprecated_at
```

5. Scientific status

Every rule is assigned:

```
ENGINEERING_PRIOR
MEASUREMENT_VERIFIED
RELIABILITY_VERIFIED
CONSTRUCT_VALIDATED
CALIBRATED
PRODUCTION_ACTIVE
DEPRECATED
```

A rule may exist in code before becoming production-authoritative.

A user-facing result SHALL NOT imply a validation status stronger than the active rule actually has.

6. Hard-gate sequence

Before estimating Dimension D , the engine SHALL execute hard gates in order.

$$G_1 \rightarrow G_2 \rightarrow \dots \rightarrow G_n$$

Recommended canonical order:

1. protocol compatibility;
2. prompt availability;
3. signal validity;
4. extractor/version compatibility;
5. sufficient observation;
6. required Evidence-family coverage;
7. required reference compatibility;
8. construct-model activation;
9. hard-confound check.

Failure behavior is rule-specific.

Possible outcomes:

```
PASS
PARTIAL
FAIL_UNRESOLVED
FAIL_SUPPRESSED
FAIL_INVALID
```

No later statistical operation may override a hard-gate failure.

7. Evidence orientation

Each Evidence Marker consumed by a Dimension rule has an orientation.

$$o_{e,d} \in \{-1, 0, +1\}$$

where:

- $+1$ supports higher expression of Dimension d ;
- -1 supports lower expression;

- 0 provides information but has no fixed directional interpretation.

Important

Orientation is **Dimension-specific**.

The same Evidence Marker may have different roles for different Dimensions.

Example:

EV_CONTEXT_RECONFIGURATION

may support higher Flexibility under one rule while serving only as contextual information for another.

No global semantic orientation is assumed.

8. Evidence strength

Each resolved Evidence Marker produces a standardized signed effect:

$$e_i$$

with associated:

$$q_i = \text{quality}$$

$$u_i = \text{uncertainty}$$

$$b_i = \text{baseline trust}$$

$$c_i = \text{confound burden.}$$

An effective Evidence contribution may be represented conceptually as:

$$\tilde{e}_i = e_i \cdot q_i \cdot b_i \cdot (1 - c_i).$$

This expression is architectural, not a permanently fixed production formula.

The exact transformation SHALL be calibrated.

9. Evidence-family independence

SoulScope SHALL distinguish:

number of measurements

from:

number of independent Evidence families.

For example:

- F0 median;
- F0 mean;
- F0 range;
- F0 variability

may provide rich Prosody information but do not count as four independent Evidence families.

Canonical families:

PRO	Prosody
ENG	Energy
TIM	Timing
PHO	Phonation
SPE	Spectral
DYN	Temporal/cross-context dynamics

Dimension rules SHALL specify the minimum independent-family coverage required.

10. Evidence coverage

For Dimension d :

$$Coverage_d = \frac{\sum_f w_f I_f}{\sum_f w_f}$$

where:

- f indexes required Evidence families;
- $I_f = 1$ if family f meets the active eligibility rule, otherwise 0;
- w_f is a versioned family-importance prior.

Until calibrated, equal weighting is preferred unless there is a strong documented scientific reason otherwise.

Hard rule

Strong evidence from one family does not compensate for a missing family when that family is explicitly required.

11. Evidence agreement

Agreement evaluates whether independently eligible Evidence families point toward compatible interpretations.

One conceptual representation is:

$$Agreement_d = 1 - \text{Conflict}(E_1, \dots, E_k).$$

The conflict function SHALL account for:

- direction disagreement;
- magnitude disagreement;
- incompatible temporal patterns;
- incompatible contextual patterns.

Agreement is not synonymous with confidence.

A weak but consistent result may have:

- high agreement;
 - low confidence.
-

12. Contradiction burden

Contradiction is stored separately from lack of support.

For Dimension d :

$$K_d = g(\text{contradictory Evidence, quality, independence, magnitude}).$$

Canonical interpretation:

$K \approx 0$	little meaningful contradiction
$K \uparrow$	stronger independent contradictory evidence

The precise scale SHALL be calibrated.

Hard principle

absence of support / contradiction. =

Missing data likewise does not count as contradiction.

13. Confound burden

Confounds SHALL be classified as:

Hard confound

Invalidates or blocks the Dimension.

Examples may include:

- incompatible extractor;
- severe clipping;
- insufficient valid speech;
- missing required task;
- microphone change invalidating comparison.

Soft confound

Reduces confidence or suppresses affected Evidence.

Examples may include:

- moderate noise;
- suspected automatic gain control;
- reported temporary voice condition;
- uncertain reference quality.

Conceptually:

$$ConfoundBurden_d = h(c_1, \dots, c_n).$$

Soft-confound effects SHALL be traceable.

14. Missingness

Missingness is represented explicitly.

Canonical missingness types:

```
STRUCTURAL_MISSING  
QUALITY_REJECTED  
PROMPT_MISSING  
REFERENCE_MISSING  
FEATURE_UNAVAILABLE  
CONFOUND_SUPPRESSED  
MODEL_NOT_VALIDATED  
PROTOCOL_UNOBSERVABLE
```

Structural missingness

The protocol cannot observe the construct.

Examples:

- Recovery;
- Reserve;
- Relational Availability.

These must not be imputed.

15. Unknown is not midpoint

SoulScope SHALL NEVER use:

`missing = 0.5.`

Nor:

`uncertain = 0.5.`

Nor:

`unresolved = balanced.`

The numeric tendency field may be absent when resolution is scientifically invalid.

Recommended representation:

```
posterior_mean = null  
posterior_lower = null  
posterior_upper = null
```

```
resolution_status = UNRESOLVED
```

when no legitimate posterior should be published.

16. Reference hierarchy

The inference engine recognizes:

```
WITHIN_PROMPT  
WITHIN_SESSION  
PERSONAL_PROVISIONAL  
PERSONAL_ESTABLISHED  
POPULATION_COMPATIBLE  
NONE
```

Reference types are not interchangeable.

A Dimension rule SHALL specify which references are:

```
REQUIRED  
PREFERRED  
OPTIONAL  
PROHIBITED
```

17. Within-session contrast

For feature or Evidence Marker x :

$$\Delta_{21} = P2 - P1$$

$$\Delta_{31} = P3 - P1$$

$$\Delta_{32} = P3 - P2.$$

These contrasts may influence inference only when:

1. source measurements are version-compatible;
2. acquisition conditions are sufficiently comparable;
3. confounds do not invalidate comparison;
4. the Dimension rule authorizes the contrast.

18. P1 interpretation

`P1_OPEN_REFERENCE` is a within-session reference condition.

It is NOT:

`NEUTRAL_EMOTION`
`HEALTHY_BASELINE`
`RESTING_BASELINE`
`TRUE_SELF`
`NORMAL_STATE`

Therefore:

$$P2 - P1$$

means:

difference from the open-reference condition,

not:

deviation from emotional neutrality.

19. Personal baseline trust

Every personal reference produces:

$$B_d \in [0, 1]$$

representing trust in that baseline for the current inference.

Baseline trust may consider:

- number of compatible historical scans;
- reference spread;
- recency;
- protocol compatibility;
- device compatibility;
- extractor compatibility;
- longitudinal stability.

The exact function is calibration-dependent.

20. Population reference

Population reference may support:

- calibration;
- measurement context;
- compatibility checks;
- research;
- bounded deviation reporting where validated.

It may not automatically replace personal reference.

A user being far from a population median does not imply dysfunction.

21. Prompt-context effect object

Every cross-prompt inference SHALL preserve:

```
contrast_id
source_prompt
target_prompt

feature_or_evidence_id

source_value
target_value
standardized_change

quality
reference_trust

active_confounds[]

interpretation_eligibility
```

22. Temporal momentum

Dimensions may carry temporal/contextual momentum:

$$M_d.$$

Momentum describes direction of movement across:

- local prompt time;
- prompt transitions;
- longitudinal scans.

It does not automatically imply improvement or deterioration.

Canonical states:

INCREASING
DECREASING
STABLE
MIXED
INSUFFICIENT

The direction is relative to the Dimension's operational scale, not to health or desirability.

23. Dimension inference classes

SoulScope defines three inference pathways.

Class D1 — Descriptive functional inference

May initially use transparent evidence aggregation.

Examples:

- Organization;
- Focus Continuity;
- Stability;
- Mobilization;
- Sustainability;
- Range.

D1 rules MAY begin as explicit engineering-rule systems, subject to validation.

Class D2 — Construct-calibrated inference

Requires a separately validated construct model.

Examples:

- Exploration;
- Processing Demand;
- Activation;
- Flexibility;
- Effort Cost;
- Openness;
- Restraint.

Before activation:

```
resolution_status = RESEARCH_ONLY
```

or:

```
resolution_status = UNRESOLVED  
reason = CONSTRUCT_MODEL_NOT_VALIDATED
```

depending on product policy.

Class D3 — Protocol-unobservable

No inference is permitted.

Examples:

- Recovery;
 - Reserve;
 - Relational Availability.
-

24. D1 provisional aggregation

For D1 Dimension d , a provisional transparent score may be constructed from family-level evidence summaries.

First derive one aggregate per Evidence family:

$$F_{d,f} = A(o_{i,d}e_i; q_i)$$

for eligible Evidence Markers i belonging to family f .

Then combine families:

$$\eta_d = \beta_{0,d} + \sum_f \beta_{d,f} F_{d,f} + \sum_{j < k} \gamma_{d,jk} I_{jk}.$$

Where:

- $\beta_{0,d}$ = intercept;
- $\beta_{d,f}$ = versioned family coefficient;
- I_{jk} = approved interaction terms;
- $\gamma_{d,jk}$ = interaction coefficients.

Then:

$$\mu_d = \sigma(\eta_d).$$

Critical status

All coefficients are:

ENGINEERING_PRIORS

until calibrated.

No hand-selected coefficient shall be described as empirically established.

25. Preferred early-stage D1 behavior

Before sufficient calibration data exist, the implementation SHOULD prefer:

- simple;
- monotonic;
- transparent;
- low-interaction

rules over complex models.

A complicated model is not scientifically superior merely because it is complicated.

26. D2 model interface

Each D2 Dimension requires an active model:

```
model_id
model_version

target_dimension
training_protocol_version

input_evidence_ids[]
input_family_requirements[]

reference_requirements[]

training_population
validation_population

calibration_method

known_limitations[]
fairness_evaluation[]

activation_status
```

The inference engine calls:

$$P(D_d \mid E, Q, B, C, P, V).$$

It SHALL return more than a point estimate.

Required output:

```
posterior_mean
posterior_lower
posterior_upper

model_confidence

input_coverage
out_of_distribution_status

feature_attribution_summary

calibration_version
```

27. Out-of-distribution behavior

A D2 model SHALL assess whether the current scan is sufficiently compatible with its validated domain.

Potential incompatibilities include:

- unsupported language;
- unsupported device;
- unsupported age range;
- unsupported acoustic characteristics;
- unsupported protocol version;
- severe task deviation.

If materially out of distribution:

```
resolution_status = UNRESOLVED  
reason = MODEL_OUT_OF_DISTRIBUTION
```

28. No LLM scoring

Generative language models SHALL NOT estimate Dimension values from:

- raw audio;
- transcripts;
- acoustic tables;
- Evidence Ledger prose.

The narrative model may receive only the completed semantic result object authorized for explanation.

29. Posterior uncertainty

For resolved Dimension d , SoulScope SHALL estimate:

$$[L_d, U_d].$$

The interval SHALL become wider when appropriate due to:

- low evidence coverage;
- low quality;
- low baseline trust;
- high contradiction;

- active soft confounds;
- weak model calibration;
- sparse personal history.

Exact interval construction depends on the active estimator.

30. Confidence is not posterior mean

SoulScope SHALL distinguish:

$$\mu_d$$

from:

$$C_d.$$

Example:

```
posterior_mean = 0.81  
confidence = 0.42
```

is valid.

This means:

the available evidence leans strongly in one direction, but the reliability of that conclusion is limited.

The system SHALL NOT collapse these two concepts.

31. Confidence components

Dimension confidence SHOULD preserve at least:

```
quality_reliability  
evidence_coverage  
evidence_agreement  
baseline_trust  
boundary_distance
```

model_calibration
confound_penalty

A provisional conceptual function is:

$$C_d = \text{Calibrate}(Q_d, E_d, A_d, B_d, D_d, M_d, 1 - K_d).$$

No simple permanent weighted average is canonically fixed in v0.1.

32. Conservative confidence aggregation

Until calibrated, confidence SHOULD behave conservatively.

A preferred family of implementations is one in which a severely weak component meaningfully limits confidence.

For example:

$$C_d \leq \min^*(Q_d, E_d, B_d)$$

where $\min^*$ represents a calibrated soft-min or conservative aggregator.

This prevents excellent signal quality from masking absent Evidence coverage.

33. Coherence

Coherence H_d represents how well independent Evidence families form a mutually compatible pattern.

High coherence requires:

- multiple independent families;
- compatible directions;
- acceptable temporal consistency;
- limited contradiction.

High coherence is not automatically positive.

It simply means the pattern is internally consistent.

34. Publication threshold

A Dimension may have a computed research estimate while still failing publication requirements.

Publication policy SHALL consider:

```
resolution_status
confidence
coverage
contradiction
construct_validation_status
active_confounds
reference_compatibility
```

The publication decision is stored separately from inference.

35. Canonical publication states

```
PUBLISHED
PUBLISHED_WITH_CAUTION
BOUNDARY_UNCERTAIN
UNRESOLVED
SUPPRESSED
RESEARCH_ONLY
INVALID
```

36. PUBLISHED

Requires:

- all hard gates passed;
 - construct activation requirements passed;
 - sufficient coverage;
 - acceptable contradiction;
 - acceptable confidence;
 - no blocking confound.
-

37. PUBLISHED_WITH_CAUTION

May be used when:

- the Dimension is validly estimable;
- uncertainty is meaningfully elevated;
- product language explicitly communicates that limitation.

The state engine may apply stricter requirements.

38. BOUNDARY_UNCERTAIN

Used when the posterior legitimately exists but lies near a semantic/state boundary.

It does not mean the underlying Dimension itself is invalid.

39. UNRESOLVED

Used when SoulScope cannot responsibly infer the Dimension.

Examples:

- insufficient evidence;
 - contradictory evidence;
 - missing required observation;
 - insufficient reference;
 - unsupported protocol;
 - unvalidated construct.
-

40. SUPPRESSED

Used when an estimate may exist internally but policy prevents publication.

Examples:

- privacy restriction;
 - temporary validation suspension;
 - fairness failure;
 - regulatory or scientific hold.
-

41. RESEARCH_ONLY

Used for investigational inference not authorized for production use.

Research-only results SHALL NOT enter user-facing Pattern or Narrative generation.

42. INVALID

Used when underlying acquisition or computation is technically invalid.

Invalid is different from unresolved.

invalid / uncertain.

43. Abstention hierarchy

When multiple abstention reasons exist, preserve all reasons but identify a primary reason.

Recommended priority:

1. INVALID_ACQUISITION
 2. PROTOCOL_UNOBSERVABLE
 3. MISSING_REQUIRED_PROMPT
 4. INSUFFICIENT_VALID_SPEECH
 5. MODEL_NOT_VALIDATED
 6. MODEL_OUT_OF_DISTRIBUTION
 7. MISSING_REQUIRED_EVIDENCE_FAMILY
 8. BASELINE_INCOMPATIBLE
 9. ACTIVE_HARD_CONFOUND
 10. HIGH_EVIDENCE_CONTRADICTION
 11. LOW_CONFIDENCE
-

44. D3 hard rules

The current production protocol SHALL always enforce:

REG-P4 Recovery
-> UNRESOLVED
-> NO_RECOVERY_COMPATIBLE_CONDITION

CAP-P2 Reserve
-> UNRESOLVED
-> NO_RESERVE_COMPATIBLE_LOAD_PROTOCOL

EXP-P4 Relational Availability
-> UNRESOLVED
-> NO_RELATIONAL_OBSERVATION

No model, state rule, Pattern rule, or narrative system may override these without a protocol-version change and Canon-compatible registry update.

45. First-scan inference

On a first scan, SoulScope may use:

- within-prompt Evidence;
- P1/P2/P3 contrasts;
- compatible population calibration where explicitly allowed.

It SHALL record:

personal_baseline_status = ABSENT

This increases uncertainty where a Dimension benefits from personal reference.

46. Longitudinal inference

When an established personal baseline exists, the engine may add:

$$z_f^{personal}$$

and:

$$\Delta_d^{longitudinal}.$$

Longitudinal change SHALL preserve:

- historical protocol version;
- extractor compatibility;
- baseline version;
- uncertainty;
- minimum detectable change where calibrated.

47. Version mismatch behavior

Semantic comparison across materially incompatible versions SHALL NOT proceed silently.

Possible outcomes:

COMPATIBLE
RECALIBRATION_REQUIRED
COMPARISON_LIMITED
COMPARISON_PROHIBITED

Historical values remain unchanged.

48. Interaction terms inside Dimension models

Interaction terms may be used when scientifically justified.

Example:

high pause load + preserved output continuity

may mean something different from:

high pause load + high fragmentation.

However, interaction terms SHALL be:

- explicitly registered;
- interpretable;
- versioned;
- validated.

The model may not discover opaque interaction logic and silently convert it into production meaning.

49. Candidate explanation object

Every resolved Dimension SHALL produce a machine-readable explanation:

```
dimension_id

result

top_supporting_evidence[]
top_contradicting_evidence[]

required_evidence_present[]
required_evidence_missing[]

reference_used

active_confounds[]

confidence_components

alternative_interpretations[]

resolution_reason
```

50. Decision Ledger architecture

Each Dimension inference creates one `DimensionDecisionRecord`.

```
decision_id
scan_id
dimension_id

rule_set_version

hard_gates[]
gate_results[]

eligible_evidence[]
ineligible_evidence[]

family_summaries[]
```

```
supporting_evidence[]
contradicting_evidence[]
missing_evidence[]

reference_inputs[]
confounds[]

model_inputs[]
model_output

posterior
confidence_components

publication_evaluation

final_status
primary_reason
secondary_reasons[]

created_at
```

51. Hard-gate record

For each gate:

```
gate_id
requirement
observed_value
threshold_or_policy_version
result
failure_consequence
```

The system SHALL NOT merely store:

```
passed = false
```

without recording why.

52. Family summary

For each required Evidence family:

```
family_id

eligible_marker_ids[]
rejected_marker_ids[]

family_effect
family_quality
family_uncertainty

direction
agreement

coverage_status
```

This lets SoulScope explain which independent Evidence families actually contributed.

53. Candidate alternatives

The Dimension engine SHALL record alternative interpretations where the active model supports them.

Example:

```
primary_candidate:
  higher Activation

alternative_candidate:
  context-specific prosodic change without sufficient
  evidence of broader Activation

reason_alternative_lost:
  multi-family evidence supported coordinated mobilization
```

For D2 constructs, alternative explanations are especially important.

54. Decision Ledger: no silent rejection

If Evidence is rejected, the ledger records:

```
evidence_id  
rejection_reason  
rejection_rule
```

Examples:

```
LOW_F0_TRACKING_CONFIDENCE  
AGC_CONFOUND  
INSUFFICIENT_PERIODICITY  
PROMPT_SCOPE_INCOMPATIBLE  
REFERENCE_MISMATCH
```

55. Decision Ledger: no silent imputation

If required Evidence is absent:

```
missing_evidence_id  
required_by_rule  
missingness_type  
downstream_effect
```

must be stored.

56. Decision Ledger: no silent override

If a lower-level estimate is overridden by a policy gate, both values must remain reconstructable.

Example:

```
research_estimate = 0.74  
  
publication_status = UNRESOLVED
```

```
reason = CONSTRUCT_MODEL_NOT_VALIDATED
```

This preserves scientific transparency.

57. Trace path

For any published Dimension:

Dimension $\leftarrow$ Inference Rule $\leftarrow$ Evidence Marker $\leftarrow$ Measurement $\leftarrow$ Audio segment.

For any unresolved Dimension:

Unresolved $\leftarrow$ failed requirement $\leftarrow$ missing/invalid Evidence or protocol limitation.

Both paths are first-class outputs.

58. Example — Focus Continuity

Illustrative only:

```
Dimension:
  COG-P3 Focus Continuity

Timing Evidence:
  pause load           -> mild contradiction
  pause density        -> mild contradiction
  speech-run duration  -> strong support
  output continuity    -> strong support

Dynamics:
  short-interval maintenance -> support

Coverage:
  sufficient

Contradiction:
  low-moderate

Result:
  posterior leans higher
```

Confidence:
moderate-high

Reason:
continuity was preserved despite somewhat greater pausing

The system does not say:

more pauses = poor focus.

It interprets the complete timing structure.

59. Example — Stability versus Flexibility

A scan may produce:

REG-P2 Stability:
HIGH

REG-P3 Flexibility:
HIGH

because:

- each prompt is internally consistent;
- the vocal pattern changes coherently when context changes.

There is no contradiction.

Likewise:

Stability = LOW
Flexibility = LOW

is possible when:

- within-prompt volatility is high;
- cross-context change lacks coherent task-related organization.

The two Dimensions are independent.

60. Example — Processing Demand abstention

Suppose:

```
Timing -> higher pause load  
Prosody -> little change  
Phonation -> unavailable  
Spectral -> contradictory
```

The engine SHALL NOT force:

```
Processing Demand = HIGH
```

Possible result:

```
status = UNRESOLVED  
  
reason =  
  INSUFFICIENT_INDEPENDENT_EVIDENCE  
  + HIGH_EVIDENCE_CONTRADICTION
```

61. Example — Restraint

SoulScope observes:

```
P1 modulation range = broad  
P2 modulation range = broad  
P3 modulation range = compressed
```

This may create research evidence for contextual modulation compression.

But if no wider range has ever been observed:

```
EXP-P3 Restraint = UNRESOLVED
```

because:

narrow range / restraint. =

Restraint requires evidence of available range plus contextual compression.

62. Model training boundary

Target labels for D2 Dimensions SHALL NOT be created merely by subjective internal annotation such as:

“this voice sounds activated.”

Validation targets must be independently justified.

Where possible, construct studies should use:

- controlled manipulation;
- validated external measures;
- repeated within-person design;
- convergent measures;
- preregistered hypotheses.

The three prompt labels themselves are not ground-truth psychological states.

63. Transcript boundary

Semantic acoustic inference SHALL NOT depend on transcript meaning unless a separate text-analysis modality is explicitly adopted.

Under the current voice-only Canon:

what was said

and:

how the qualified voice behaved acoustically

remain separate.

A transcript may support task-compliance checks where authorized, but it SHALL NOT silently inject semantic-content interpretation into Constellation scoring.

64. Fairness boundary

No Dimension model becomes production-active without evaluation for material performance differences across relevant groups and contexts.

Evaluation may include:

- language;
- accent;
- age;
- sex-linked voice characteristics;
- device;
- environment;
- speech style.

Observed subgroup differences do not automatically imply unfairness.

The question is whether measurement reliability, calibration, abstention, and error rates remain acceptable.

65. Calibration set separation

Training, tuning/calibration, and final evaluation data SHALL be separated at the participant level.

Repeated scans from one participant SHALL NOT be split across training and final evaluation in a manner that creates identity leakage.

66. Calibration targets

At minimum, calibration should evaluate:

- posterior calibration;
 - interval coverage;
 - reliability;
 - abstention accuracy;
 - subgroup calibration;
 - out-of-distribution detection;
 - stability across protocol-compatible repeat scans.
-

67. Threshold governance

Every numerical publication threshold SHALL have:

```
threshold_id
threshold_version

target_dimension

value
unit_or_scale

scientific_status
calibration_dataset
validation_dataset

activation_date
```

No threshold may exist only as an undocumented constant in code.

68. Rule change classes

PATCH

Bug fix that does not change scientific meaning.

MINOR

Adds compatible rule behavior without changing existing interpretation.

MAJOR

Changes:

- semantic direction;
- required Evidence;
- estimator;
- reference behavior;
- abstention behavior;
- publication meaning.

Major changes require new historical compatibility rules.

69. Reproducibility contract

Given identical:

- Measurement Record;
- Evidence Ledger;
- rule versions;
- model versions;
- calibration versions;

the semantic inference SHALL reproduce the same machine result within declared deterministic numerical tolerance.

Narrative wording may vary only within the separately authorized Narrative layer.

70. Acceptance-test examples

INF-UNK-001

Given a Dimension with missing required Evidence family:

Expected: UNRESOLVED .

Never midpoint.

INF-D3-001

Given current three-prompt protocol:

Expected: REG-P4 Recovery = UNRESOLVED .

INF-D3-002

Given current three-prompt protocol:

Expected: CAP-P2 Reserve = UNRESOLVED .

INF-D3-003

Given current three-prompt protocol:

Expected: EXP-P4 Relational Availability = UNRESOLVED.

INF-CON-001

Given strong support and strong independent contradiction:

Expected: confidence decreases and/or result becomes unresolved according to active policy.

Contradiction must not disappear through averaging.

INF-REF-001

Given absent personal baseline:

Expected: personal deviation absent.

Population reference must not substitute silently.

INF-LLM-001

Given raw audio provided to narrative model:

Expected: architecture violation.

Narrative must receive completed Semantic Result only.

INF-VER-001

Given incompatible rule/model version during longitudinal comparison:

Expected: comparison blocked or explicitly limited.

Historical result remains unchanged.

71. What v0.1 freezes

The following are canonical now:

- rule taxonomy;
 - hard-gate architecture;
 - Evidence-family independence;
 - explicit missingness;
 - contradiction as first-class information;
 - separate baseline trust;
 - separate confidence and posterior tendency;
 - D1/D2/D3 inference classes;
 - D3 hard abstentions;
 - no LLM scoring;
 - no direct feature-to-emotion rule;
 - explicit publication layer;
 - complete Decision Ledger;
 - versioned thresholds;
 - versioned models;
 - reconstructable trace path.
-

72. What v0.1 intentionally does not freeze

The following remain calibration-dependent:

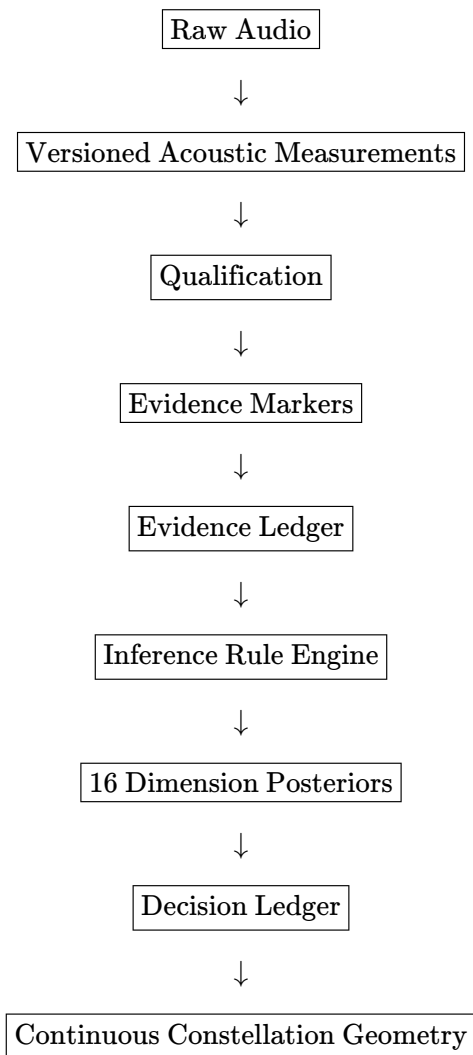
- coefficients;
- feature-family weights;
- precise posterior mapping;
- interval estimator;
- confidence formula;
- contradiction cutoff;
- minimum publication confidence;
- state boundaries;
- blend margin;
- model architecture;
- D2 construct targets.

This is intentional.

Freezing those values before SoulScope has its own validation data would create false precision.

73. Canonical semantic pipeline

The complete semantic backend is now:



State reasoning occurs only after this point.

74. Final inference law

SoulScope does not ask whether enough numbers exist to produce an answer.

It asks whether enough independent, qualified, compatible evidence exists to justify one.

When the answer is no:

Unresolved is the correct result.